POWER ENTRY (5-pin XLR from external +-17 V PSU; reverse-polarity Schottkys; star point)
DISTRIBUTION (PTC per wing per rail; local branches VA audio, VP phones, VM meters, VL logic)
MOTHERLOOM (master in the centre: CN1 = left wing, CN2 = right wing; Delta pinout)
GROUND CANCEL BUFFER (feeds every re-inverter; summing amps use the raw GCANCEL line)
MIX L: summing amp (GCANCEL-referenced), re-inverter, master insert
MIX R: summing amp (GCANCEL-referenced), re-inverter, master insert
MASTER FADER (dual 100 mm A10K, pin 1 = GNDA) + post-amps +10 dB
MONO: inverting sum of POST L+R, re-inverter
MAIN OUTPUTS: DRV134 -> 10u NP in the Sense loop -> mute relay (NO) -> XLR male, pin 2 hot
POWER-ON MUTE: rails > ~14 V for ~2 s closes all five relays; either rail low opens them at once
SUBGROUP 1: sum, re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL
SUBGROUP 2: sum, re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL
SUBGROUP 3: sum, re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL
SUBGROUP 4: sum, re-invert, insert, fader + post-amp, DRV134 out, pan to MIX, AFL
AUX MASTER 1: OPA1656 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL
AUX MASTER 2: OPA1656 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL
AUX MASTER 3: OPA1656 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL
AUX MASTER 4: OPA1656 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL
AUX MASTER 5: OPA1656 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL
AUX MASTER 6: OPA1656 sum (22K || 100p) + re-invert, level, buffer, DRV134 out, AFL
STEREO RETURN 1: receivers (R normalled to L), level, buffers, MIX / 1-2 / 3-4, PFL
STEREO RETURN 2: receivers (R normalled to L), level, buffers, MIX / 1-2 / 3-4, PFL
STEREO RETURN 3: receivers (R normalled to L), level, buffers, MIX / 1-2 / 3-4, PFL
STEREO RETURN 4: receivers (R normalled to L), level, buffers, MIX / 1-2 / 3-4, PFL
EXT INPUT (2-track / DAW) and MIX / EXT monitor select (SW latched = EXT)
PFL / AFL: OPA1656 sum (22K || 100p) + re-invert, SOLO LEVEL, follower
PFL DETECT (1.5 V per press vs 0.74 V), complementary ramps, rude-solo flasher
PFL SWITCHER: series J112 (Delta ED2780 sh.1) -> CRS L/R followers
CONTROL ROOM: C/ROOM dual A10K, followers, DRV134, mute relays, TRS outs
HEADPHONES: PHONES dual A10K, OPA1612 + LME49600 composite (+9.9 dB), 2R2 out, PHONESGND
REMOTE METER HEADER (for a meter bridge)
METER L: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings
METER R: full-wave rectifier, precision peak / average, follower (TL072), 2 dB ladder, 2 x 10-LED strings

Nets joined by labels; refs auto-assigned
Generated by tools/gen_master.py from master_section_spec.md v0.5

Sheet: /
File: master.kicad_sch

Title: Master section — goa console

Size: User
Date: 2026-10-19
Rev: 0.5
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